

EEHPCWG Workshop 2025

Session: Water management for system and facility cooling

PGW25 Cooling Fluid Figure of Merit

Steve Dean, Distinguished Technologist

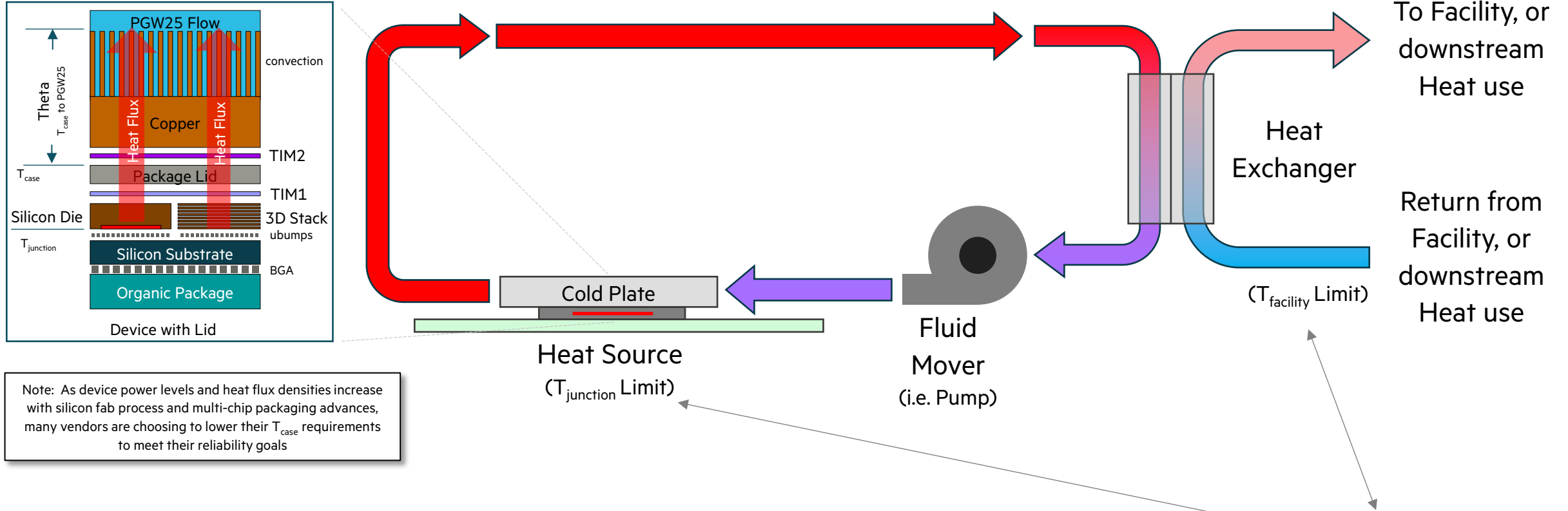
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HPE

EEHPC WG workshop, June 2025



Cooling Fluid Function: Enabling the Capture and Movement of Heat Energy

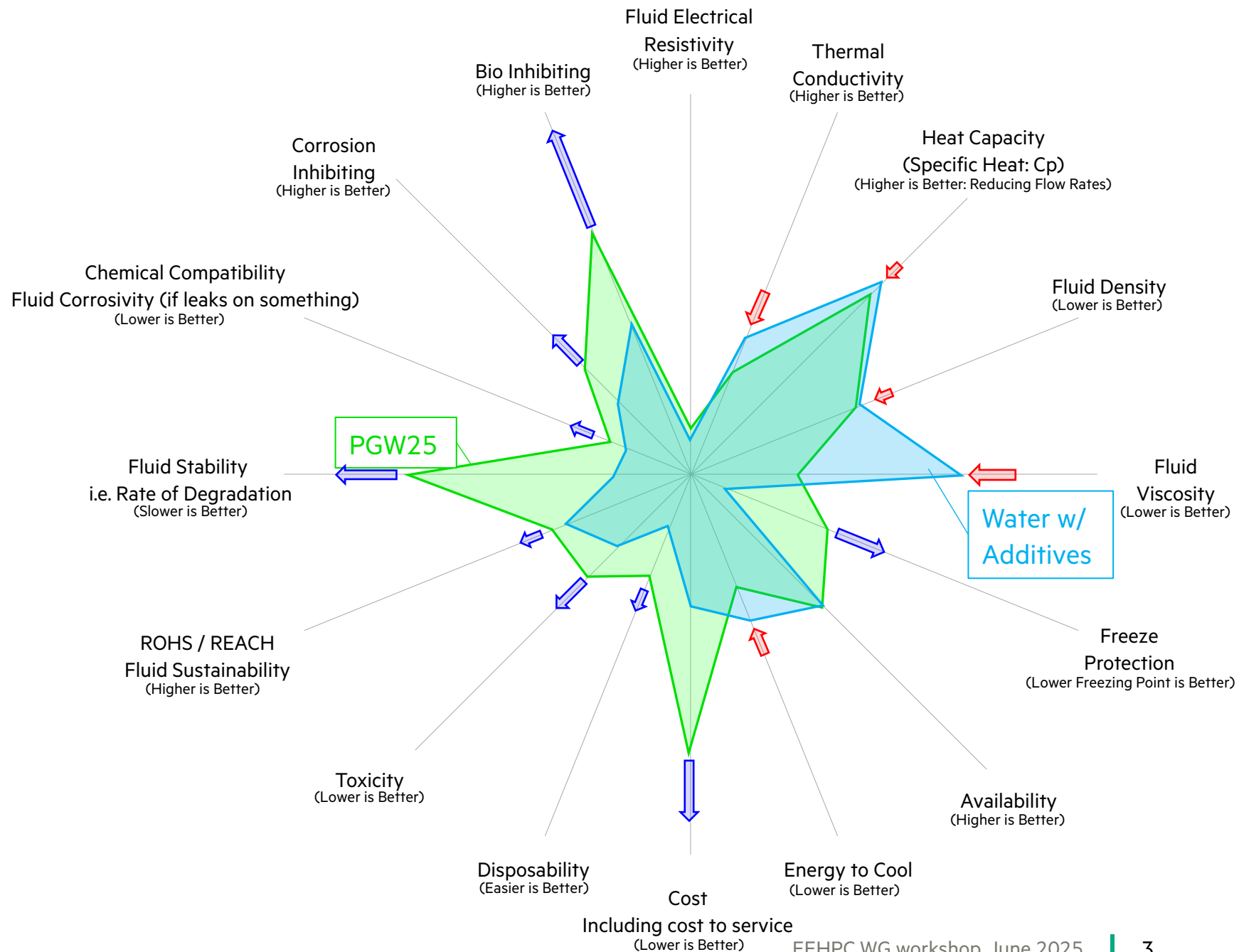


Note: Solutions must exist within a Cooling Temperature Window (W) of: $W = T_{\text{junction}} - T_{\text{facility}}$

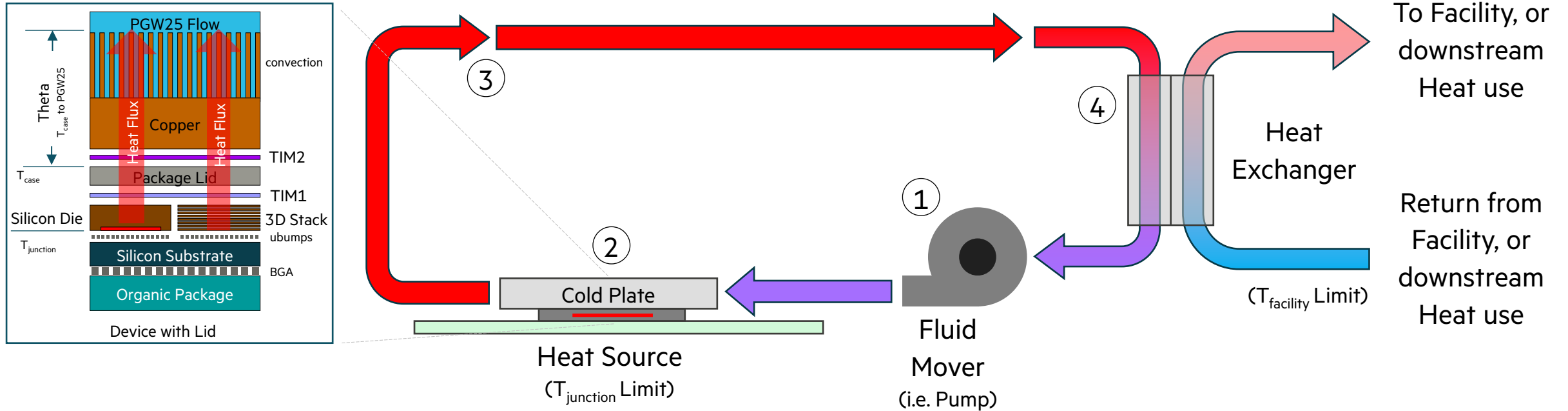
Adopting PGW25 vs Water w/Additives

PGW25 vs Water Trade-offs:

- Reduction of some physical properties:
 - Overall lower heat transfer capabilities
- Increased stability and reliability:
 - Bio-Inhibiting
 - Fluid Stability
 - Cost (including cost to service)
 - Very low toxicity (food grade)
- Explicitly designing future systems to optimize for cooling fluid choice



Targeting Areas of Cooling Fluid Improvement



Research Focus Areas

1. Reduce energy required for moving the cooling fluid.
2. Improve effectiveness of heat capture into the cooling fluid.
3. Improve heat moved by the cooling fluid.
4. Support downstream use of heat produced.

The Amphoteric Nature of Water

Water is naturally amphoteric.

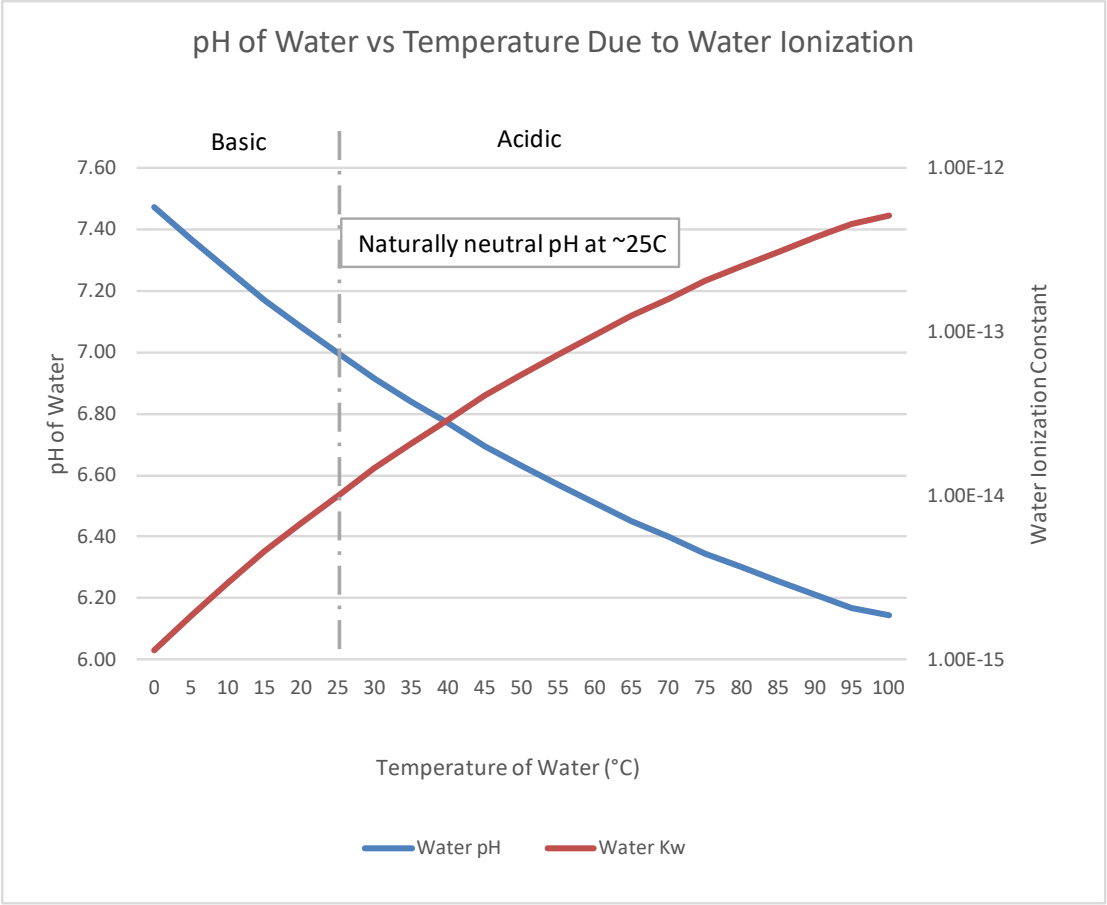
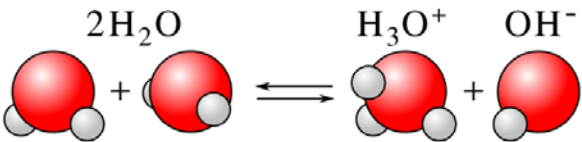
This means that water naturally goes through self ionization where water molecules (H₂O) can donate or accept protons to form hydronium ions, H₃O⁺ (a strong acid) and hydroxide ions, OH⁻ (a strong base).

The self ionization equilibrium constant (i.e. the rate at which this occurs) is Kw.

This equilibrium constant is temperature dependent and it increases with temperature.

So as water is heated, it's self ionization rate increases resulting in the water shifting from a basic pH at low temperatures to an acidic pH at higher temperatures.

Temp (°C)	Water pH	Water Kw
0	7.47	1.14E-15
5	7.37	1.86E-15
10	7.27	2.93E-15
15	7.17	4.57E-15
20	7.08	6.81E-15
25	7.00	1.01E-14
30	6.92	1.47E-14
35	6.84	2.09E-14
40	6.77	2.92E-14
45	6.69	4.07E-14
50	6.63	5.48E-14
55	6.57	7.24E-14
60	6.51	9.55E-14
65	6.45	1.26E-13
70	6.40	1.59E-13
75	6.34	2.04E-13
80	6.30	2.51E-13
85	6.26	3.09E-13
90	6.21	3.80E-13
95	6.17	4.57E-13
100	6.14	5.13E-13



Thank you

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